

ASSIGNMENT # 01

Machine Learning

Submission Deadline: 7th Oct 2026

Air Quality Monitoring

Air pollution is a major environmental concern worldwide, with pollutants such as carbon monoxide (CO), nitrogen oxides (NO_x, NO₂), non-methane hydrocarbons (NMHC), and benzene (C₆H₆) having significant impacts on human health.

The Air Quality Dataset (collected in an Italian city from March 2004 to February 2005) provides hourly averaged responses of several chemical sensors, along with ground truth (GT) pollutant concentrations and meteorological variables.

This dataset allows us to explore how well sensor signals and weather conditions can predict the actual concentration of air pollutants.

Dataset Description

Each row in the dataset corresponds to one hourly measurement.

Main Variables:

- **Date, Time** → Timestamp of measurement.

Ground Truth Pollutants (possible targets for prediction):

- CO(GT) → True hourly averaged CO concentration (mg/m³).
- NMHC(GT) → True hourly averaged Non-Methane Hydrocarbons (µg/m³).
- C₆H₆(GT) → True hourly averaged Benzene (µg/m³).
- NO_x(GT) → True hourly averaged Nitrogen Oxides (ppb).
- NO₂(GT) → True hourly averaged NO₂ (µg/m³).

Sensor Signals (features):

- PT08.S1(CO) → CO-sensitive metal oxide sensor signal.
- PT08.S2(NMHC) → NMHC-sensitive metal oxide sensor signal.
- PT08.S3(NO_x) → NO_x-sensitive metal oxide sensor signal.
- PT08.S4(NO₂) → NO₂-sensitive metal oxide sensor signal.
- PT08.S5(O₃) → O₃-sensitive metal oxide sensor signal.

Environmental Conditions (features):

- T → Temperature (°C).
- RH → Relative Humidity (%).
- AH → Absolute Humidity.

*Further Details of dataset can be acquired from this page: <https://archive.ics.uci.edu/dataset/360/air+quality>

Assignment Task

Data Preparation

- Load the dataset into Python (CSV provided).
- Handle missing values
- Choose one pollutant as your **target variable**
- Select relevant features (sensors + weather conditions).
- Split dataset into training (80%) and testing (20%).

Model Training

- Train a **Regression Model from scratch** (Linear and Polynomial (of appropriate degree: Explain how you obtained that appropriate degree) to predict the chosen pollutant. Once Implemented from scratch, cross verify your results by using built in command in sklearn library to fit the data. Compare the two results. Record training and test performance for each case.
- Train a **Decision Tree Regressor** to predict the chosen pollutant. Experiment with different values of max_depth (e.g., 3, 5, 10, None). Record training and test performance for each case.

Model Evaluation

- Compute and report for all models:
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
- Plot performance metrics vs. epoch (For regression) and Performance metric vs tree depth for Decision tree.

Model Visualization

- Plot the decision tree for a small depth (e.g., max_depth=3).
- Display feature importances: Which sensor/environment variable contributes most to predictions?

Baseline Comparison

- Compare Linear Regressor performance with the Decision Tree Regressor.
- Discuss differences in capturing linear vs. non-linear patterns.

Discussion (short written answers)

- Which features were most important for predicting your chosen pollutant?
- Which performed better overall: Decision Tree or Linear Regression? Why?

Deliverables

1. Submit a Jupyter Notebook (.ipynb) or .py file containing:
 - Data preprocessing and handling of missing values.
 - Decision Tree model training with different depths.
 - Evaluation metrics and plots.
 - Visualizations (tree diagram, feature importances).
 - Linear Regression comparison.
 - Short discussion
2. A pdf document with code and all reported results and discussions.

Note: Your submitted code should be neat and clean with proper comments added.
